Citation intent extraction and application to question answering using Prior knowledge graph

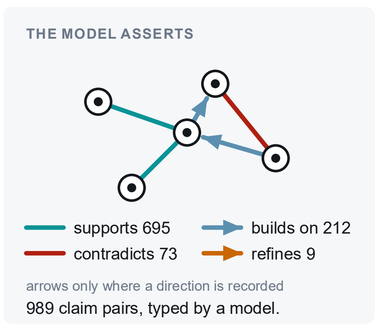
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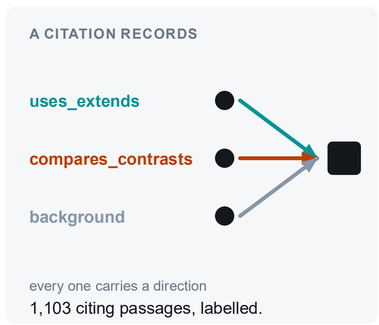
0) Overview:

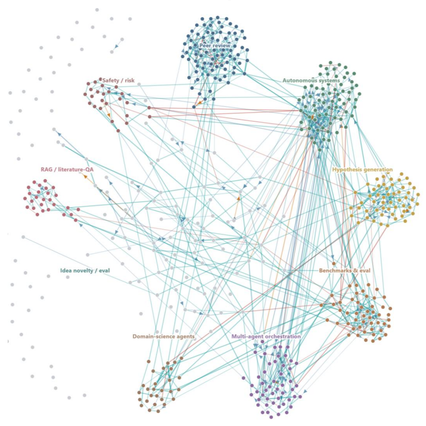
152 papers ▶ Semantic graph ▶ Mine citations ▶ Classify citation intent ▶ Enrich graph ▶ Graph QA agent ▶ Evaluate answers

1) Introduction:

Prior ingests a corpus of research papers on a topic and builds a map of them. It pulls out each paper's contributions, and draws typed, directed relations between them, so you can ask "what is actually known about X?" and get an answer grounded in the literature rather than a plausible sounding summary. The graph was missing any citation information. The first task was to figure out how best to use citations to enrich the graph, and then to answer questions using a graph exploration agent. Test corpus: 152 papers on "agents for the scientific process".







2) Citation labelling:

The citations between papers in the corpus are mined and then given labels. The main label is intent, for which we define a taxonomy, to characterize the reasons why one paper might cite another:

- **Background:** cited as context, prior art or a passing mention
- **Uses/extends:** citing paper actually adopts or builds on the target.
- **Compares/contrasts**: citing paper sets its own work specifically against the target

A judge was then developed to assign these labels to our mined citations and the evidence package was tuned: claim passage + cited paper abstract were chosen to strike the best balance between cost and accuracy. An interesting finding during the prompt tuning was that telling the model *what a class is* leads to far better classification than telling it *what a neighbouring class isn't* when trying to enforce a tie-breaker. E.g. tell the judge to classify grouped field critiques as background, rather than not to classify them as compares/contrasts.

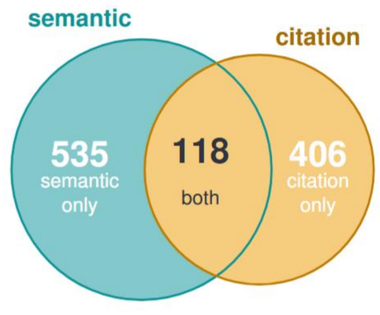
3) Validation:

A human labelled gold set of 80 random citations was created and the labels agreed 95% of the time on this set (macro-F1 0.892) . Opus 5, a stronger model, was also given a separate random subset of 100 citations, the same context package, but only the original simple definitions rather than the engineered prompt, in order to check faithfulness to those definitions. It showed substantial agreement with a Cohen's kappa of 0.79. The disagreements were then investigated and found to fall on a just a couple of known definition boundaries. (e.g. group critiques, where a paper is listed among others to convey a limitation of the field)

BLIND GOLD SET
95.0%
76 of 80 sites I labelled by hand
95% CI 87.8 to 98.0

BLIND SECOND JUDGE
0.79
Cohen's kappa against Opus 5
100 sites, clean-room rubric

4) Citation/semantic overlap



THE 118 SHARED PAIRS: WHAT THE MODEL CALLED THEM

	supports	builds_on	contradicts
background	57	23	8
uses_extends	10	9	0
compares_contrasts	19	10	4

The citation graph was found to have quite a narrow overlap with the original semantic graph, (both are rolled up to paper pairs, strongest relation dominates). In such a fast moving field as this, papers often don't have the time to cite each other even when related. The 118 overlapping pairs were investigated, and largely it was found that the information was complementary, both the citation edge and the semantic edge provided useful information, often covering different facets of the relationship between the papers.

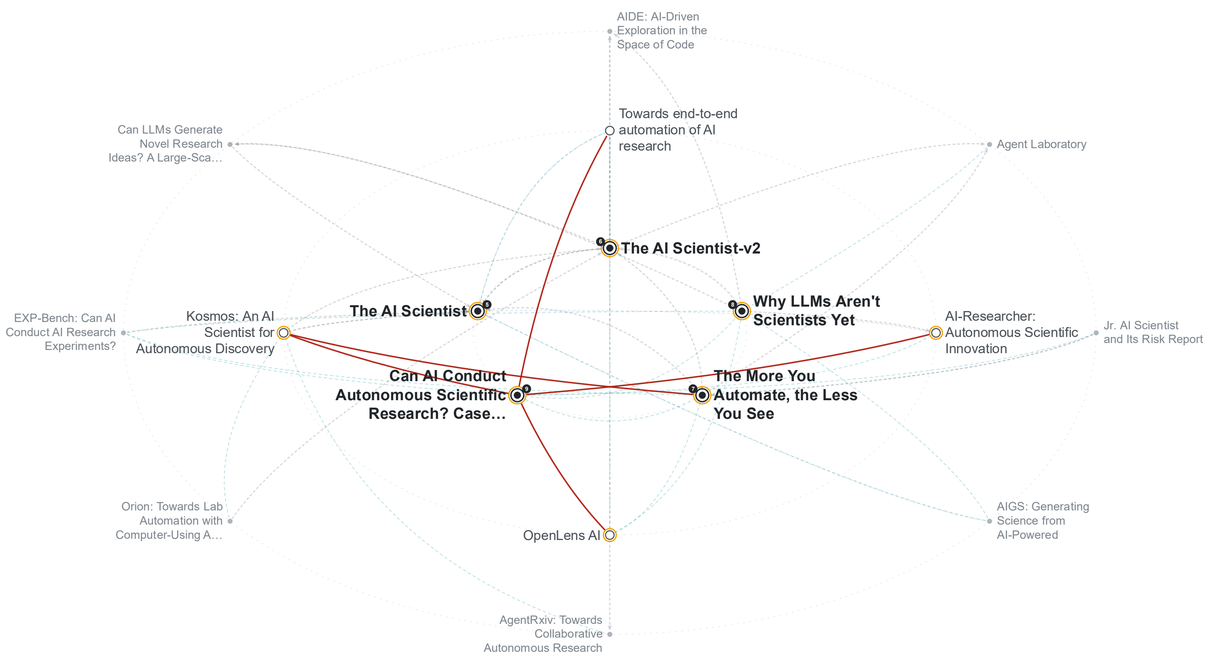
5) Using graph for question answering

With the fully enriched graph complete, we evaluated its effectiveness for question answering. We tested a sample of eight questions (6 structural, 2 control) across three agent configurations: a web-based agent, a graph-based agent (that explores over our graph but not the web), and a baseline "null arm" lacking both capabilities. A more capable model then acted as a judge, scoring the responses on groundedness, correctness, and usefulness to select a preferred answer. Across the six structural questions, the judge preferred the graph-based agent's answer five out of six times, and these verdicts remained stable across 2 order-swapped re-evaluations. Mean scores are shown in the charts below. Out of corpus control questions were also given to the graph arm as a test, and it successfully identified its own limitations in these cases: *"I was not able to find coverage of this topic in the material available to me" .*

During this process, the graph agent leaves a trace of its tool usage, which can be visualized as a subgraph; two sample questions demonstrating this are shown below and to the right.

6) Exploration visualisation Q1: Tensions

My PI thinks fully autonomous AI scientists already produce publishable research. Is that contested in the literature?

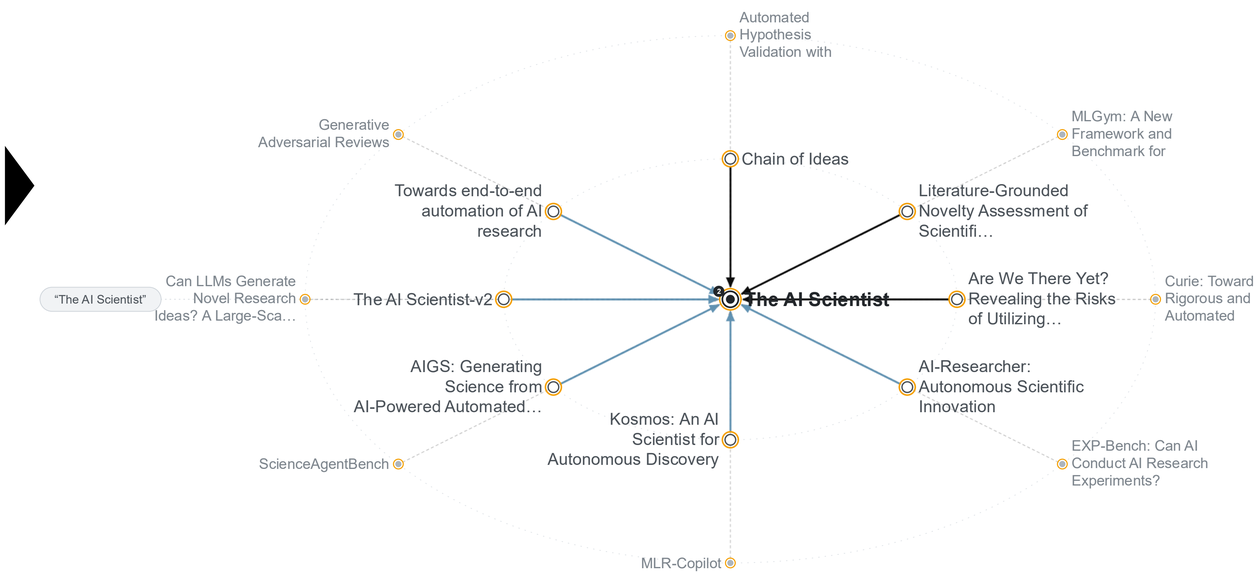


WHAT IT DID, IN ORDER

1. topic map "autonomous AI scientist generating re…"
2. keyword search "AI scientist publishable research aut…"
3. look up by name "Why LLMs Aren't Scientists Yet"
4. look up by name "Can AI Conduct Autonomous Scientific…"
5. neighbourhood: The AI Scientist
6. neighbourhood: The AI Scientist-v2
7. neighbourhood: The More You Automate, the Le…
8. neighbourhood: Why LLMs Aren't Scientists Yet
9. neighbourhood: Can AI Conduct Autonomous Sci…
10. read justifications: Can AI Conduct Autonomous Sci… · contradicts
11. read justifications: The More You Automate, the Le… · contradicts

7) Exploration visualisation Q2: Builds on

I'm writing a related-work section on The AI Scientist (Lu et al.). Which later papers actually extend its method, as opposed to just citing it in passing?



WHAT IT DID, IN ORDER

1. look up by name "The AI Scientist"
2. neighbourhood: The AI Scientist
3. read justifications: The AI Scientist · builds_on
4. citation detail: The AI Scientist → Chain of Ideas
5. citation detail: The AI Scientist → Literature-Grounded Nov…
6. citation detail: The AI Scientist → Are We There Yet? Revea…

HOW THE AGENT ENGAGED

- chose it — named as a call argument (n = step)
- read its evidence — justification pulled
- listed only — delivered, not read
- named in the final answer
- a search the agent typed (dotted lines → what it found)
- "no match" — nothing in the corpus fit it

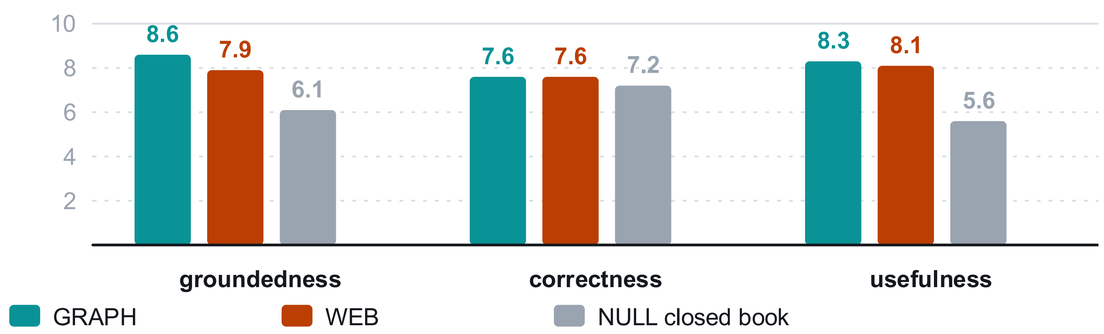
RELATION

- contradicts (symmetric)
- refines
- builds on
- supports (symmetric)
- cites (citation layer)

contradicts and supports are recorded symmetric — no direction is claimed

solid = read · dashed = listed only

THE SIX STRUCTURAL QUESTIONS · MEAN OF 3 ORDER-SWAPPED JUDGE RUNS



+2.5 the atlas gains over the closed-book arm. Web search gains +1.8.

2 / 2 controls lost to web, as predicted.

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